

Serg Nesterov

Lead Engineer, Payments & Billing

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SUMMARY

I build and run the systems that take money: payment gateway orchestration, recurring billing and dunning, fraud and chargeback defense. I lead development at a four-brand consumer fintech, responsible for everything technical and setting direction for a team of four. Twelve years working remotely with US companies, and at my best in a small one, where an engineer can own a whole system instead of a limited slice of it.

EXPERIENCE

Credit Glory, Lead Developer

November 2022 – Present

Technical lead for a four-brand consumer fintech. Responsible for everything technical, setting direction for a team of four developers and working with leadership on where the business goes next.

- Own multi-gateway payment routing and failover across five processors (NMI, Payarc, Nexio, Pathway, Spreedly), covering MID-level routing, max-ticket handling and forced-gateway operations. Built the stored-credential (CIT/MIT) charge flows and Account Updater card synchronization.
- Designed the subscription, invoice and cart state machines behind recurring billing, then extended them with a 90-day dunning sequence, guarantee and discount-plan interactions, and refund handling on retired MIDs.
- Integrated Kount for fraud detection across all four brands, including Device Data Collection fingerprinting and Consumer Insights. Built chargeback defense on top of it: RDR, CDRN and Ethoca alert-window handling, with alert-to-transaction matching by ARN julian-date decoding, worked against Visa VAMP and TC40 thresholds.
- Built a browser softphone from scratch on Twilio Voice and TaskRouter. Conference internals, warm and cold transfer through the Participants API, dual-channel recording, and task-version fences that make out-of-order webhooks idempotent.
- Ran Twilio Flex across four brands: IVR, queue and skill topology sync, agent presence and idle state, worker audits, and a Spanish-language queue rollout.
- Rebuilt the transactional email stack on Mailgun, moving 180+ templates over and adding webhook-driven suppression and unsubscribe handling.
- Built credit-report ingestion and dispute automation against bureau portals using Selenium behind residential proxies.
- Ran production infrastructure for four brands on DigitalOcean: Docker and Traefik, a four-instance Grafana fleet, OOM and container debugging, and per-brand runbooks. Tuned PostgreSQL query performance down to JIT cost and EXPLAIN (ANALYZE, BUFFERS) on large buffer-resident tables.

DebugBear, Senior Full-Stack Engineer

December 2021 – October 2022

www.debugbear.com

Web performance monitoring product. Worked directly with the founder to expand it, and was the second-highest contributor to the codebase with 299 commits.

- Built the public Free Test tool: URL input, in-progress and results views, test history, a prerendered landing page, conversion into ongoing monitoring, and an internal admin view.
- Moved chart and third-party metric queries onto ClickHouse, then built comparison ranges and configurable aggregation periods on top.
- Worked in browser and performance internals: netlog capture across a breaking Chrome and Puppeteer upgrade, Lighthouse version upgrades and score-magnitude bugs, request-initiator snippet extraction, and resource-hint validation that flags preloaded async and defer resources.

- Implemented subscription and billing logic on Chargebee, including plan-limit enforcement on downgrades and trial-upgrade invoicing.

Credit Glory, Senior Full-Stack Developer

December 2019 – November 2021

First tenure. Shipped customer-facing product features and internal operational tooling across a Rails, PostgreSQL and Sidekiq codebase, and built the payment gateway integrations and recurring-billing flows that became the foundation of the multi-gateway platform.

Groove, Full Stack Developer, then Senior Frontend Developer

March 2014 – November 2019

www.groovehq.com

Remote US SaaS company building a shared inbox, support widget and knowledge base. Started on the marketing site and custom Rails CMS, then moved onto the product applications.

- Part of the team that rewrote a server-rendered Rails shared inbox into a React progressive web app backed by a Rails API and a GraphQL layer. Developer velocity and application responsiveness both improved, and it became the platform for later apps.
- Pushed for and implemented Storybook and a company design system, which unified how the applications looked and let developers iterate faster.
- Member and unofficial lead of the Frontend Working Group, responsible for major frontend decisions and for promoting best practices.

Freelance, Web Designer and Developer

January 2012 – March 2014

OPEN SOURCE

nixpkgs

github.com/NixOS/nixpkgs

Maintainer, listed in [maintainers/maintainer-list.nix](https://maintainers.nixos.org/maintainer-list). Long-time Nix and NixOS user for reproducible development and deployment environments.

sanctuary, jsverify

github.com/sanctuary-js/sanctuary

Contributor to typed functional programming and property-based testing libraries in JavaScript.

SquareFace

Operate a multi-server Minecraft network on Hetzner: Java plugin development, Velocity proxy, Rust tooling, MariaDB and self-hosted monitoring.

SKILLS

Payments & Billing (*Expert*): Payment orchestration, Gateway routing and failover, PSP integration, Stored credentials (CIT/MIT), Account Updater, Tokenization, Authorization, capture, refunds, Recurring billing, Dunning, Reconciliation

Fraud & Chargeback Risk (*Expert*): Kount, Device fingerprinting, Chargeback and dispute management, RDR, CDRN, Ethoca, Visa VAMP, TC40, ARN matching

Backend (*Expert*): Ruby, Ruby on Rails, PostgreSQL, Sidekiq, GraphQL, Webhook idempotency, State machines, TypeScript, JavaScript

Telephony & Comms (*Advanced*): Twilio Voice, Twilio TaskRouter, Twilio Flex, WebRTC softphone, Call control and transfer, IVR and queue routing, Mailgun

Infrastructure (*Advanced*): Linux, Docker, Traefik, DigitalOcean, Grafana, CI/CD, Nix, Selenium, Playwright

Also: Go, Elixir, Rust, React, UI/UX design

LANGUAGES

English (*Fluent*), German (*Working Knowledge*), Russian (*Native Speaker*)

INTERESTS

Recreational Mathematics, Theory of Computation, Photography